

EDUCATIVE COMMENTARY ON ISI 2019 MATHEMATICS PAPERS

(Last Revised on 17/09/2020)

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Introduction

Commentaries of ISI B.Stat. Entrance Tests began in 2016 because it was the year in which some exciting questions came to the notice of the author. Later commentaries from 2014 were added, especially at the request of some students who found that unlike the JEE, there are no sources to get the answers.

Unfortunately, this year there are very few good problems and so the exercise was more an annual ritual.

As in the other commentaries, unless otherwise stated, all the references are to the author's *Educative JEE Mathematics*, published by Universities Press, Hyderabad.

I am thankful to the persons who made the question papers available to me as well as those who pointed out errors in the earlier draft. The elegant solution to Q.1 of Paper 1 at the end was contributed by Faiz Akhtar.

Readers are invited to send their comments and point out errors, if any, in the commentary. They may be sent either by e-mail (kdjoshi314@gmail.com) or by an SMS or a WhatsApp message on mobile (9819961036).

ISI BStat-BMath-UGA-2019 Paper 1 with Comments

N.B. Each question has four options of which ONLY ONE is correct. There are 4 marks for a correctly answered, 0 marks for an incorrectly answered question and 1 mark for each unattempted question.

Q.1 Let $P(X) = X^4 + a_3X^3 + a_2X^2 + a_1X + a_0$ be a polynomial in X with real coefficients. Assume that

$$P(0) = 1, P(1) = 2, P(2) = 3, \text{ and } P(3) = 4.$$

Then, the value of $P(4)$ is

- (A) 5 (B) 24 (C) 29 (D) not determinable from the given data.

Answer and Comments: (C). The most tempting answer is 5. Mercifully it is given as an option. So, it cannot be correct! The most straightforward way to find the right answer is to determine the polynomial $P(X)$, i.e. the numbers a_0, a_1, a_2, a_3 from the data and then evaluate $P(4)$ by a direct substitution. For this we need four equations in the four unknowns a_0, a_1, a_2 and a_3 . From the given values, they come out to be

$$a_0 = 1 \tag{1}$$

$$a_0 + a_1 + a_2 + a_3 = 2 \tag{2}$$

$$a_0 + 2a_1 + 4a_2 + 8a_3 = 3 \tag{3}$$

$$\text{and } a_0 + 3a_1 + 9a_2 + 27a_3 = 4 \tag{4}$$

Using (1), the remaining three equations simplify to

$$a_1 + a_2 + a_3 = 0 \tag{5}$$

$$a_1 + 2a_2 + 4a_3 = -7 \tag{6}$$

$$\text{and } a_1 + 3a_2 + 9a_3 = -26 \tag{7}$$

Because of (5), the other two equations simplify to

$$a_2 + 3a_3 = -7 \tag{8}$$

$$a_2 + 4a_3 = -13 \tag{9}$$

which can be solved to give $a_3 = -6, a_2 = 11$ and finally, $a_1 = -5$. Hence by a direct calculation,

$$P(4) = 256 - 6 \times 64 + 11 \times 16 - 5 \times 4 + 1 = 177 - 148 = 29 \quad (10)$$

A straightforward problem. The computations are too simple to warrant any alternatives. Those who are familiar with solving a system of linear equations by using what is called **row reduction** will recognise that in the present problem, that is effectively what we have done.

But there are a few aspects of the problem itself which deserve some educative comments. Instead of arriving at the system of equations (5)-(7), we could have merely put $a_0 = 1$ into (2)-(4) and recast them in a matrix form as

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 8 \\ 3 & 9 & 27 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -14 \\ -78 \end{bmatrix} \quad (11)$$

In effect, we have not divided the second and the third row by 2 and 3 respectively. The advantage of resisting this temptation is that here the coefficient matrix is an example of a well-known matrix, called the **Vandermonde matrix** for whose inverse there is a simple formula. More generally, for any real numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ if we define $V(\alpha_1, \alpha_2, \dots, \alpha_n)$ as

$$V(\alpha_1, \alpha_2, \dots, \alpha_n) = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ \alpha_1 & \alpha_2 & \alpha_3 & \dots & \alpha_n \\ \alpha_1^2 & \alpha_2^2 & \alpha_3^2 & \dots & \alpha_n^2 \\ \vdots & & & & \vdots \\ \alpha_1^{n-1} & \alpha_2^{n-1} & \alpha_3^{n-1} & \dots & \alpha_n^{n-1} \end{bmatrix} \quad (12)$$

then

$$|V(\alpha_1, \alpha_2, \dots, \alpha_n)| = \prod_{1 \leq i < j \leq n} (\alpha_j - \alpha_i) \quad (13)$$

where the R.H.S. is a product of $\frac{n(n-1)}{2}$ factors. An interesting corollary is that $|V(\alpha_1, \alpha_2, \dots, \alpha_n)| \neq 0$ if and only if all the α_i 's are distinct.

(What is even more interesting is that in most of the applications, it is only this corollary, and not the full formula (13), that is needed. A direct inductive proof of the corollary, which bypasses (13) is available.)

In our problem, $n = 3$, $\alpha_1 = 1$, $\alpha_2 = 2$ and $\alpha_3 = 3$. So the coefficient matrix in (11) has determinant $(2 - 1) \times (3 - 1) \times (3 - 2) = 2 \neq 0$. (Again, for this the Corollary is enough.) Hence (11) has a unique solution. This does not help much in finding the solution. But if the question was merely whether option (D) is correct, we know the answer. (The values of a_2, a_3, a_4 can be found by Cramer's rule. But that is not much of a simplification.)

Another comment is regarding how a tempting line of attack is *not* applicable in the present problem. There is a well known result, called **Lagrange's interpolation** which says that for any positive integer n , given distinct $\alpha_1, \alpha_2, \dots, \alpha_n$ and (not necessarily distinct) $\beta_1, \beta_2, \dots, \beta_n$ there is a unique polynomial $P(x)$ of degree $n - 1$ or less which assumes the value β_i at α_i for $i = 1, 2, \dots, n$. This can be proved using Vandermonde matrices again. An explicit formula for this interpolating polynomial can also be given. But we refrain from it because it is irrelevant in the present problem, as the interpolating polynomial would come out to be simply $x + 1$ (which is also easy to guess by common sense, because then the problem would reduce to a newspaper puzzle of finding the next term of a sequence). The catch is that the question already stipulates that $P(x)$ is a polynomial of degree 4 and so Lagrange's formula is doomed to fail. (Incidentally, the reason the formula is called 'interpolation' is that it is used to approximate an (unknown) function $f(x)$ whose values at a few points $\alpha_1, \alpha_2, \dots, \alpha_n$ are known by some method, possibly an experiment. Although the polynomial $P(x)$ it generates makes sense for any real x , the approximation is generally applied only for values of x that lie inside the smallest interval containing $\alpha_1, \alpha_2, \dots, \alpha_n$. Outside this interval, the error, that is, the difference $|f(x) - P(x)|$ can be quite unpredictable. When a formula is applied to values outside the interval spanned by the data, it is called **extrapolation**.)

Nevertheless, if we use the (incorrect) guess $X + 1$ for $p(X)$ as a correcting term, we get a surprisingly short, albeit tricky solution suggested by Faiz Akhtar. Let $q(X) = p(X) - (X + 1)$. Then $q(X)$ is a polynomial of degree 4 in X . The data implies that 0, 1, 2 and 3

are roots of $q(X)$. This determines $q(X)$ uniquely as $q(X) = X(X - 1)(X - 2)(X - 3)$ and hence $p(X) = X(X - 1)(X - 2)(X - 3) + X + 1$. So, $p(4) = 24 + 4 + 1 = 29$.

Q.2 Let f be a real-valued, differentiable function defined on the real line $\mathbb{R}$ such that its derivative f' is zero at exactly two distinct real numbers α and β . Then,

- (A) α and β are points of local maxima of the function f .
- (B) α and β are points of local minima of the function f .
- (C) one must be a point of local maximum and the other a point of local minimum of the function f .
- (D) given data is insufficient to conclude about either of them being a local extrema points.

Answer and Comments: (D). Assume that $f'(\alpha) = f'(\beta) = 0$ and further that $\alpha < \beta$ without loss of generality. Then f is one-to-one on each of the intervals $(-\infty, \alpha)$, (α, β) and (β, ∞) . For otherwise, $f(x_1) = f(x_2)$ would imply, by Lagrange's Mean Value Theorem (LMVT) that f' vanishes at some point other than α and β . A continuous, one-to-one function on any interval must be either strictly increasing or decreasing by the Intermediate Value Property (IVP). So we conclude that f is strictly increasing/decreasing on each of these three intervals. There will be a local maximum or minimum at α , if f changes its behaviour from increasing to decreasing (or vice versa) as we move from the interval $(-\infty, \alpha)$ to the interval (α, β) . An analogous assertion holds for β .

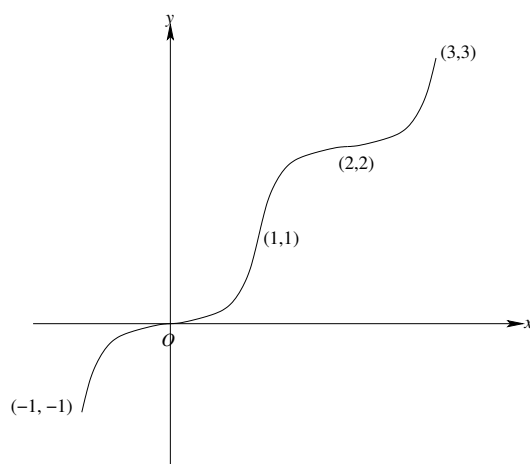
But it may happen that there is no change of behaviour at a zero of f' . A standard example is when $f(x) = x^3$. 0 is the only zero of $f'(x) = 3x^2$. But f is increasing on both $(-\infty, 0)$ and $(0, \infty)$. (The explanation here is that 0 is a double root of the derivative f' . So $f'(x)$ does not change its sign as x passes through 0. More generally, if α is a root of even multiplicity of f' , then there is neither a local maximum nor a local minimum of $f(x)$ at α .)

In the given problem, if both α and β are double roots of f' , then f will have neither a local maximum nor a local minimum. An

actual example where this happens can be constructed by modifying the example just given. Thus define $f(x)$ by

$$f(x) = \begin{cases} x^3, & x \leq 1 \\ (x-2)^3 + 2, & x \geq 1 \end{cases}$$

Then f' has two zeros, viz. 0 and 2. But it is strictly increasing on entire $\mathbb{R}$. The graph of $f(x)$ is shown below.



Q.3 A school allowed the students of a class to go to swim during the days March 11 to March 15, 2019. The minimum number of students the class should have had that ensures that at least two of them went to swim on the same set of dates is:

- (A) 6 (B) 32 (C) 33 (D) 121

Answer and Comments: (C). This is one of those problems where the real challenge is more to translate a real life data into a mathematical one. The single word ‘set’ gives the clue. Let S be the set of the students in the class and D the set of dates from 11 th to 15 th March 2019. We may take D as the set $\{11, 12, 13, 14, 15\}$. For every student $x \in S$, let D_x be the set of those dates on which x went to swim. Then D_x is some subset of D (possibly, the empty subset if x never went to swim on any of these five dates). In all there are 2^5 , i.e. 32 subsets of D . If S has more than 32 students, then it is bound to happen

that $D_x = D_y$ for some $x, y \in S$, with $x \neq y$. That means that both x and y went to swim on the same set of dates. On the other hand if S has 32 or fewer students then we can arbitrarily select as many distinct subsets of D and assign them to the students. So, 33 is the least number of students in the class that will ensure that at least two (distinct) students went to swim on the same dates.

For those familiar with the concept of a power set, let $P(D)$ be the set of all subsets of D (including the empty subset and the entire set D). It is easy to show that the number of elements in $P(D)$ is $2^{|D|}$ where $|D|$ is the number of elements in the set D . For this reason, $P(D)$ is called the **power set** of D . The argument above amounts to defining a function $f : S \rightarrow P(D)$ by $f(x) = D_x$. When the domain set S has more elements than the codomain set $P(D)$, there cannot be any one-to-one function from S to $P(D)$.

Q.4 Let $a_1 < a_2 < a_3 < a_4$ be positive integers such that

$$\sum_{i=1}^4 \frac{1}{a_i} = \frac{11}{6}.$$

Then $a_4 - a_2$ equals

- (A) 11 (B) 10 (C) 9 (D) 8.

Answer and Comments: (B). We have

$$\frac{1}{a_1} > \frac{1}{a_2} > \frac{1}{a_3} > \frac{1}{a_4} \quad (1)$$

Also the average of these four numbers is $\frac{11}{24}$. Therefore $\frac{1}{a_1} > \frac{11}{24}$, which implies $a_1 < \frac{24}{11}$. Since a_1 is a positive integer, it can only be 1 or 2. We discuss these two cases separately.

- (i) $a_1 = 1$. In this case, $\frac{1}{a_2} + \frac{1}{a_3} + \frac{1}{a_4} = \frac{11}{6} - 1 = \frac{5}{6}$. The average of these three numbers is $\frac{5}{18}$. Hence we have $\frac{1}{a_2} > \frac{5}{18}$ which gives $a_2 < \frac{18}{5}$. Therefore a_2 equals 2 or 3. The first possibility gives $\frac{1}{a_3} + \frac{1}{a_4} = \frac{5}{6} - \frac{1}{2} = \frac{1}{3}$. So, once again, $\frac{1}{a_3} > \frac{1}{6}$ which means $a_3 = 3, 4$ or 5 . But we also have $3(a_3 + a_4) = a_3 a_4$ which rules out both $a_3 = 3$ and $a_3 = 5$ since a_4 is a positive integer. The

only possibility left is when $a_3 = 4$. In this case $\frac{1}{a_4} = \frac{1}{12}$. So, the integers $a_1 = 1, a_2 = 2, a_3 = 4$ and $a_4 = 12$ do satisfy the conditions of the problem. And in this case at least we have $a_4 - a_2 = 10$.

As only one of the options is correct, and we have found one situation when (B) holds, we might as well stop here in a multiple choice test. But to finish the solution honestly, we must also consider the possibility $a_1 = 1, a_2 = 3$. In this case $\frac{1}{a_3} + \frac{1}{a_4} = \frac{5}{6} - \frac{1}{3} = \frac{1}{2}$. The average is $\frac{1}{4}$ which forces $a_3 < 4$ which is impossible since $a_2 = 3$ and $a_3 > a_2$.

- (ii) $a_1 = 2$. In this case $\frac{1}{a_2} + \frac{1}{a_3} + \frac{1}{a_4} = \frac{11}{6} - \frac{1}{2} = \frac{4}{3}$. Hence $\frac{1}{a_2} > \frac{4}{9}$ which implies a_2 equals 1 or 2. Both are ruled out since $a_2 > a_1$.

Summing up, $(1, 2, 4, 12)$ is the only quadruple of integers which satisfies the data. The key idea of the solution is that when distinct numbers are added, the largest one must exceed the average. After this, it is rather tedious to consider all the possibilities. Those who guess this quadruple stand to win by saving time.

- Q.5 Three children and two adults want to cross river using a row boat. The boat can carry no more than a single adult or, in case there is no adult in the boat, a maximum of two children. The least number of times the boat needs to cross the river to transport all five people is:

(A) 9 (B) 11 (C) 13 (D) 15.

Answer and Comments: (B). This is a slight variation of a classic puzzle in which there are two adults and only two children. Common sense is the best approach. The crucial task is to get the two adults across. There is no point in an adult bringing the boat back. Nor is there any point in a single child rowing the boat forward except possibly at the end. Note also that the last forward trip cannot be by an adult, because for this to be possible, somebody else must have brought the boat backward in the last but one trip, and this person will not be able to go with the adult in the last forward trip.

These observations serve to design a scheme of rowing and also show that it is optimal, i.e. has least number of trips. Call the two adults as A and B and the three children as C, D and E . It is foolish to begin

with only one person in the very first trip because then he/she will also have to bring the boat back and thus the first two trips are a waste. So, first send two children, say C and D , across and let C bring the boat back. Then A can go across and D will bring the boat back. So a minimum of four crossings is needed to get an adult across and bring the boat back. After two such sessions of four trips each for A and B , only C, D, E will be left. Any two of them can go forward, either one of them can bring the boat backward and take the third child with him/her in the last forward trip. Hence the minimum number of trips needed is $4 + 4 + 3 = 11$.

Q.6 Let M be a 3×3 matrix with all entries being 0 or 1. Then, all possible values for $\det(M)$ are

- (A) $\{0, \pm 1\}$ (B) $\{0, \pm 1, \pm 2\}$ (C) $\{0, \pm 1, \pm 3\}$ (D) $\{0, \pm 1, \pm 2, \pm 3\}$.

Answer and Comments: (B). If a matrix has determinant d , by interchanging any two rows of it we get a matrix with determinant $-d$. So we may remove the $\pm$ sign from all the options and worry if there are any matrices of the given description which have 0, 1, 2 and 3 as determinants.

Trivially, the zero matrix and the identity matrix I_3 have determinants 0 and 1 respectively. A 3×3 matrix with entries only 0 or 1 having determinant 2 can be constructed by trial, e.g. the matrix

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

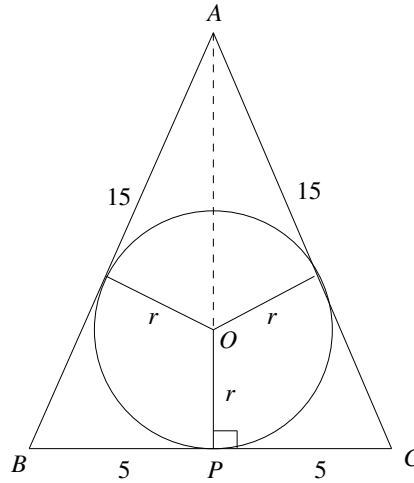
This reduces the problem to either finding a 3×3 matrix M with only 0 and 1 as possible entries and determinant 3 or else showing that no such matrix exists. We claim that there is no such matrix. As there are only 2^9 distinct 3×3 matrices with 0 and 1 as possible entries, we can list all of them and try one-by-one till we either get a desired matrix or till we have exhausted all of them. But this is impracticable as $2^9 = 512$ is too large a number for it. We look for better ways

to prove the non-existence. Suppose $M = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ is a matrix with only 0 or 1 as entries. Note that every 2×2 submatrix of M

has determinant 0 or ± 1 . So, if some row has at least one 0, then expanding the determinant w.r.t. that row we see that it can be at most 2. Hence for $\det(M)$ to equal 3, all rows must have only 1's. But then $\det(M) = 0$. So there is no matrix M of the given type whose determinant is 3. Hence (C) and (D) are incorrect options.

The question is of the same spirit as (but simpler than) Q.8 in Paper 2 of JEE 2018 Advanced, about the maximum possible value of a 3×3 determinant with 0, 1 and -1 as possible entries.

- Q.7 In the following picture, ABC is an isosceles triangle with an inscribed circle with centre O . Let P be the mid-point of BC . If $AB = AC = 15$ and $BC = 10$, then OP equals:



- (A) $\frac{\sqrt{5}}{\sqrt{2}}$ (B) $\frac{5}{\sqrt{2}}$ (C) $2\sqrt{5}$ (D) $5\sqrt{2}$.

Answer and Comments: (B). Denote the length of OP by r . In the usual terminology, O is the incentre and r is the in-radius of $\triangle ABC$. From the data, the semi-perimeter s equals $\frac{15+15+10}{2} = 20$. Hence Δ , the area of the triangle equals $20r$. But the area also equals $\frac{1}{2}hBC$ where h is the altitude through A . As $AB = AC$, $AP \perp BC$. From the right-angled triangle APC , we have $h = \sqrt{(15)^2 - 5^2} = \sqrt{200} = 10\sqrt{2}$. Hence $\Delta = 50\sqrt{2}$. Equating this with $20r$ gives $r = \frac{5}{\sqrt{2}}$.

A very simple problem once the idea of equating two expressions for the area strikes. The computations are also easy and not prone to numerical errors.

Q.8 For every real number $x \neq -1$, let $f(x) = \frac{x}{x+1}$. Write $f_1(x) = f(x)$ and for $n \geq 2$, $f_n(x) = f(f_{n-1}(x))$. Then,

$$f_1(-2)f_2(-2)\dots f_n(-2)$$

must equal

$$(A) \frac{2^n}{1.3.5.\dots(2n-1)} \quad (B) 1 \quad (C) \frac{1}{2}\binom{2n}{n} \quad (D) \binom{2n}{n}.$$

Answer and Comments: (A). For small values of n , $f_n(-2)$ can be found by direct computation. Thus,

$$\begin{aligned} f_1(-2) &= \frac{-2}{-2+1} = 2 \\ f_2(-2) &= f(f_1(-2)) = f(2) = \frac{2}{3} \\ f_3(-2) &= f(f_2(-2)) = f\left(\frac{2}{3}\right) = \frac{2/3}{5/3} = \frac{2}{5} \\ f_4(-2) &= f(f_3(-2)) = f\left(\frac{2}{5}\right) = \frac{2/5}{7/5} = \frac{2}{7} \\ f_5(-2) &= f(f_4(-2)) = f\left(\frac{2}{7}\right) = \frac{2/7}{9/7} = \frac{2}{9} \end{aligned}$$

The pattern is clear, viz.

$$f_n(-2) = \frac{2}{2n-1} \tag{1}$$

This can be proved easily by induction on n . Indeed,

$$\begin{aligned} f_{k+1}(-2) &= f(f_k(-2)) = f\left(\frac{2}{2k-1}\right) \\ &= \frac{\frac{2}{2k-1}}{\frac{2}{2k-1} + 1} \\ &= \frac{2}{2k+1} \end{aligned} \tag{2}$$

which proves the inductive step.

By a direct computation, the product $f_1(-2)f_2(-2)\dots f_n(-2)$ equals $\frac{2 \times 2 \times \dots \times 2}{1.3.5 \dots (2n-1)} = \frac{2^n}{1.3.5 \dots (2n-1)}$. Hence (A) holds.

A very straightforward problem. In an MCQ, the inductive proof of (1) can be skipped.

Q.9 Let the integers a_i for $0 \leq i \leq 54$ be defined by the equation

$$(1 + X + X^2)^{27} = a_0 + a_1X + a_2X^2 + \dots + a_{54}X^{54}.$$

Then $a_0 + a_3 + a_6 + \dots + a_{54}$ equals

$$(A) 3^{26} \quad (B) 3^{27} \quad (C) 3^{28} \quad (D) 3^{29}.$$

Answer and Comments: (A). Call each side of the given equation as $p(X)$. It is a polynomial in X of degree 54. It would be horrendous to calculate all the coefficients $a_0, a_1, \dots, a_{54}$ individually and then add the ones we want, viz. $a_0 + a_3 + a_6 + \dots + a_{54}$. But there is a short cut for it if we notice that the suffixes are the terms of an A.P. with common difference 3. For any polynomial $q(X) = b_0 + b_1X + b_2X^2 + \dots + b_{n-1}X^{n-1} + b_nX^n$, it is well known that

$$b_0 + b_1 + b_2 + \dots + b_n = q(1) \quad (1)$$

A less well known result is that

$$b_0 + b_2 + b_4 + \dots + b_{2r} + \dots = \frac{q(1) + q(-1)}{2} \quad (2)$$

where the sum on the L.H.S. ends with b_n if n is even and with b_{n-1} if n is odd. This can be proved by noting that

$$b_0 - b_1 + b_2 - b_3 + \dots + (-1)^r b_r + \dots + (-1)^n b_n = q(-1) \quad (3)$$

(2) follows by adding (1) and (3) and dividing by 2.

The crucial role played by -1 here is that all its even powers are 1 and the odd powers are -1 . And this happens because it is a square root of 1, different from 1.

If instead of a square root, we let ω be a cube root of 1, other than 1 itself, then ω^2 is also a cube root. The factorisation of $x^3 - 1$ as $(x - 1)(x - \omega)(x - \omega^2)$ implies

$$1 + \omega + \omega^2 = 0 \quad (4)$$

Further, $\omega^r = \omega^s$ if and only if $r - s$ is divisible by 3. In particular, $\omega^r = 1$ if and only if 3 divides r . Similar assertions hold for powers of ω^2 .

The analogous result of (2) is

$$b_0 + b_3 + b_6 + \dots + b_{3r} + \dots = \frac{q(1) + q(\omega) + q(\omega^2)}{3} \quad (5)$$

which is proved by adding the expressions for $q(1)$, $q(\omega)$ and $q(\omega^2)$ and using (4).

With this spadework, the problem can be tackled easily. We have

$$a_0 + a_3 + a_6 + \dots + a_{54} = \frac{p(1) + p(\omega) + p(\omega^2)}{3} \quad (6)$$

Since $p(X) = (1 + X + X^2)^{27}$, $p(\omega) = 0$ by (1). Also $1 + \omega^2 + (\omega^2)^2 = 1 + \omega^2 + \omega^4 = 1 + \omega + \omega^2 = 0$ which implies that $p(\omega^2)$ vanishes too. On the other hand $p(1) = (1 + 1 + 1)^{27} = 3^{27}$. Putting these values in (6), we get $a_0 + a_3 + a_6 + \dots + a_{54} = \frac{3^{27}}{3} = 3^{26}$.

A good problem. The technique can be generalised. As a complex number, ω equals $e^{2\pi i/3}$. More generally, for any positive integer r , if we let $\alpha = e^{2\pi i/r}$, then $\alpha, \alpha^2, \dots, \alpha^{r-1}$ are all r -th roots of unity (i.e. their r -th powers equal 1). They add up to -1 . Further,

$$b_0 + b_r + b_{2r} + \dots + b_{kr} + \dots = \frac{q(1) + q(\alpha) + q(\alpha^2) + \dots + q(\alpha^{r-1})}{r} \quad (7)$$

which is proved analogously to (5) and can be applied in some counting problems. For example, taking $r = 4$ (so that $\alpha = i$), one can count the number of six digit integers (with no leading zeros allowed) whose digital sum is divisible by 4. The answer is 225002.

Q.10 An examination has 20 questions. For each question, the marks that can be obtained are either $-1, 0$ or 4 . Let S be the set of possible total marks that a student can score in the examination. Then the number of elements in S is

- (A) 93 (B) 94 (C) 95 (D) 96.

Answer and Comments: (C). The maximum and the minimum possible scores are 80 and -20 respectively. So S is a subset of the set of all integers from -20 to 80. This set has 101 elements (including 0). But not all these scores are attainable. For example, a score of 79 cannot be attained as we shall see. So to see how many elements there are in S , we have to subtract from 101 the number of unattainable scores.

Let x, y, z denote, respectively, the numbers of questions on which 4, -1 and 0 marks are scored. Then the total score attained is $4x - y$. Here the variables x, y, z are non-negative integers from 0 to 20 and constrained by the condition

$$x + y + z = 20 \tag{1}$$

Since $z \geq 0$, we can paraphrase the constraints as $x \geq 0, y \geq 0$ and $x + y \leq 20$. Certainly, every score m from -20 to 0 is attainable with $x = 0, y = -m$. (Note that $m \leq 0$.) There is also no difficulty in attaining any score m from 1 to 68. We let k be the least multiple of 4 greater than or equal to m . Then $\frac{k}{4}$ is some integer between 0 and 17. Further $k - m$ is an integer between 0 to 3. So taking $x = \frac{k}{4}$ and $y = k - m$, we get $4x - y = k - (k - m) = m$. This shows that $m \in S$ since the constraint $x + y \leq 20$ is also satisfied because $x \leq 17$.

Trouble arises for integers greater than 68. Starting at the upper end, the score 79 is unattainable. To achieve it, x will have to be bigger than 19. But x cannot be bigger than 20. If $x = 20$, then y and z have to be 0. So, the scores 79, 78 and 77 are unattainable. 75 is attainable as we see by taking $x = 19$ and $y = 1$. But 74 is not, because for $4x - y = 74$ to hold, x will have to be at least 19 and y at least 2, a contradiction. Similarly, $73 \notin S$.

Below 72, 71 and 70 are in S . For 71, we take $x = 18, y = 1$, while for 70, we take $x = 18, y = 2$. The constraint $x + y \leq 20$ holds in

both the cases. However, 69 is not attainable, because to express it as $4x - y$, would require $x \geq 18$ and $y \geq 3$, violating the constraint.

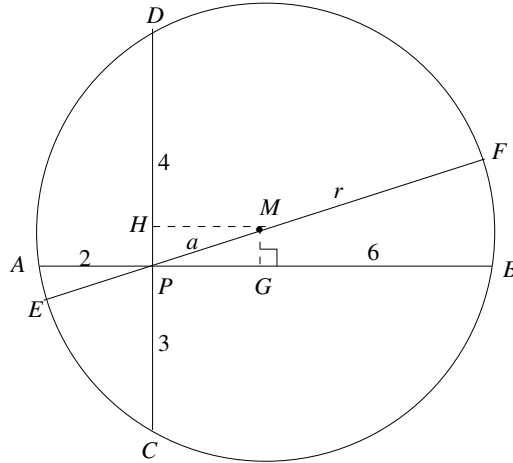
Summing up, out of the 101 integers from -20 to 80 , all integers except $79, 78, 77, 74, 73$ and 69 are in S . Hence $|S| = 101 - 6 = 95$.

A good problem, requiring simple but novel thinking.

- Q.11 Chords AB and CD of a circle intersect at right angle at the point P . If the lengths of AP, PB, CP, PD are $2, 6, 3, 4$ units respectively, then the radius of the circle is:

(A) 4 (B) $\frac{\sqrt{65}}{2}$ (C) $\frac{\sqrt{66}}{2}$ (D) $\frac{\sqrt{67}}{2}$.

Answer and Comments: (B). The paper-setters have been kind by specifying the length of the segment PD as 4. There is a theorem that for any fixed point P inside a circle, for any variable chord XY passing through P , the product $XP.PY$ is a constant. So we could have calculated PD as $\frac{AP.PB}{CP}$ which comes out to be $\frac{12}{3}$. (The constant $XP.PY$ is often called the power of P w.r.t. the circle.)



To find r , the radius of the circle, we let M be its centre and a its distance from P . Then applying the theorem just mentioned to the diametrical chord EF through P , we get

$$12 = EP.PF = (EM - PM).(PM + MF) = (r - a)(r + a) = r^2 - a^2 \quad (1)$$

So we would know r if we can find a , i.e. PM . For this, drop perpendiculars MG, MH from M to the chords AB and CD respectively. Then G and H are the midpoints of AB and CD respectively. So

$$PG = AG - AP = 4 - 2 = 2 \quad (2)$$

$$\text{and similarly, } PH = PD - HD = 4 - \frac{7}{2} = \frac{1}{2} \quad (3)$$

Hence from the right angled triangle PGM ,

$$a^2 = PM^2 = PG^2 + GM^2 = PG^2 + PH^2 = 4 + \frac{1}{4} = \frac{17}{4} \quad (4)$$

Putting this into (1) gives $r^2 = 12 + \frac{17}{4} = \frac{65}{4}$. Hence $r = \frac{\sqrt{65}}{2}$.

A simple problem once the key idea of the power of a point strikes.

Q.12 The locus of a point (x, y) in the plane satisfying $\sin^2 x + \sin^2 y = 1$ consists of

- (A) A circle that is centred at the origin.
- (B) infinitely many circles that are all centred at the origin.
- (C) infinitely many lines with slope ± 1 .
- (D) finitely many lines with slope ± 1 .

Answer and Comments: (C). Normally, in a locus problem, a moving point (x, y) satisfies certain conditions from which we have to find an equation satisfied by x and y . In the present problem, on the other hand, one such equation is given, viz.,

$$\sin^2 x + \sin^2 y = 1 \quad (1)$$

and we have to find what type of subset of the plane is specified by it. So in this sense the problem is unusual and the word ‘locus’ is somewhat misleading. It would have been better to introduce a set, say S , as $\{(x, y) \in \mathbb{R}^2 : \sin^2 x + \sin^2 y = 1\}$ and then ask to identify the nature of S .

Anyway, as the problem stands, from the identity $\sin^2 x + \cos^2 x = 1$ and (1) together we get

$$\sin y = \pm \cos x \quad (2)$$

On the other hand, combining (1) with $\sin^2 y + \cos^2 y = 1$, we also get

$$\cos y = \pm \sin x \quad (3)$$

Dividing (2) by (3)

$$\tan y = \pm \cot x = \pm \tan\left(\frac{\pi}{2} - x\right) \quad (4)$$

Taking the positive sign gives

$$y = \frac{\pi}{2} - x + n\pi \quad (5)$$

where n is any integer. For each n , this is a straight line with slope -1 .

On the other hand, taking the negative sign in (4) gives

$$y = x - \frac{\pi}{2} + n\pi \quad (6)$$

where n is any integer. For each n , this is a straight line with slope 1 .

Put together, the given locus consists of infinitely many lines with slopes ± 1 .

The problem is more on trigonometric equations than coordinate geometry. Instead of going through (3), the answer can also be obtained directly from (2) by writing $\cos x$ as $\sin(\frac{\pi}{2} - x)$.

Q.13 The number of integers $n \geq 10$ such that the product $\binom{n}{10} \binom{n+1}{10}$ is a perfect square is

(A) 0 (B) 1 (C) 2 (D) 3.

Answer and Comments: (B). Let A, B denote $\binom{n}{10}$ and $\binom{n+1}{10}$ respectively. Then A, B are some positive integers with

$$n! = A(n-10)!(10)! \quad (1)$$

$$\text{and } (n+1)! = B(n-9)!(10)! \quad (2)$$

Multiplying the two, we get

$$(n+1)(n!)^2 = AB((n-10)!)^2((10)!)^2(n-9) \quad (3)$$

i.e.

$$(n+1)(n-9) = \frac{AB((n-10)!)^2((10)!)^2(n-9)^2}{(n!)^2} \quad (4)$$

Since AB is a perfect square by assumption, we see that the R.H.S. is the square of a rational number. But since the L.H.S. is an integer, this rational number must also be an integer. Thus we get that $(n+1)(n-9) = y^2$ for some integer y . Put $x = n - 9$. Then we get

$$x(x+10) = y^2 \quad (5)$$

where x, y are integers. Thus the problem reduces to finding integral solutions of (5). Putting $z = x + 5$, (5) can be rewritten as

$$z^2 = y^2 + 25 \quad (6)$$

Hence $(z+y)(z-y) = 25$. The only way this is possible is when $z+y = 25$ and $z-y = 1$. This gives $z = 13, y = 12$. Hence $x = z-5 = 8$ and finally $n = x + 9 = 17$. So we get that for $n = 17$ and for no other value of n , AB is a perfect square. So (B) holds.

An excellent problem requiring only elementary number theory. The substitutions made are all bijective and hence reversible. So it is not necessary to verify that $n = 17$ indeed satisfies the condition. Still, a direct calculation gives, after cancellation of common factors, that $\binom{17}{10} = \binom{17}{7} = 17 \times 13 \times 11 \times 2$ and similarly $\binom{18}{10} = \binom{18}{8} = 18 \times 17 \times 13 \times 11$. So their product is indeed the square of $17 \times 13 \times 11 \times 6$.

Q.14 Let $a \geq b \geq c \geq 0$ be integers such that $2^a + 2^b - 2^c = 144$. Then $a + b - c$ equals:

- (A) 7 (B) 8 (C) 9 (D) 10.

Answer and Comments: (B). Dividing throughout by 2^c we get

$$2^{a-c} + 2^{b-c} = 1 + 2^{4-c} \quad (1)$$

As the L.H.S. is an integer, so is the R.H.S. Hence $0 \leq c \leq 4$. When c is given the values 0, 1, 2, 3 and 4, the R.H.S. becomes 145, 73, 37, 19 and 10 respectively. Out of these only 10 can be expressed as a sum of

two powers of 2, viz. $8 = 2^3 + 2^1$. So we must have $c = 4$, $a - c = 3$ and $b - c = 1$. This gives $a = 7$ and $b = 5$. Hence $a + b - c = 7 + 5 - 4 = 8$.

Not a very inspiring problem, in sharp contrast with the last one.

- Q.15 The number of integers n for which the cubic equation $X^3 - X + n = 0$ has three distinct integer solutions is:

(A) 0 (B) 1 (C) 2 (D) infinite.

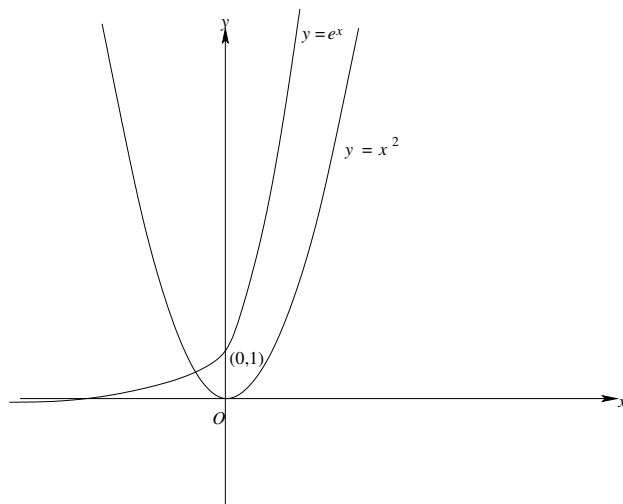
Answer and Comments: (B). Rewrite the given equation as $(X - 1)X(X + 1) = -n$. Then $-n$ is a product of three consecutive integers, of which the middle one can have three distinct values. This is possible only when $n = 0$ and $X = 0, 1$ or -1 . So there is only one value of the integer n (viz. $n = 0$) which satisfies the condition in the problem.

An even more dull problem than the last one.

- Q.16 The number of real solutions of the equation $x^2 = e^x$ is:

(A) 0 (B) 1 (C) 2 (D) infinite.

Answer and Comments: (B). The equation equates an algebraic expression (viz., x^2) with an exponential one, (viz., e^x). Usually, there is no way to solve such equations, except by trial. But in the present problem we want only the number of solutions and not the actual solutions. And this can be done by calculus. In effect, the problem asks to find the number of points of intersection of the two graphs $y = x^2$ and $y = e^x$. As both the graphs are very standard, this is a workable proposition. We show both these graphs in the figure below.



It is clear that the two graphs intersect at only one point.

For an analytical solution, we consider the function $f(x) = x^2 - e^x$. As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$. But $f(0) = 0 - 1 < 0$. So f has at least one zero in $(-\infty, 0)$. But there cannot be more than one zero in $(-\infty, 0)$ because $f(x)$ is strictly decreasing in it since $f'(x) = 2x - e^x < 0$ for all $x < 0$. For $x \geq 0$, e^x increases far more rapidly than x^2 . So f cannot vanish anywhere in $(0, \infty)$.

Q.17 The number of distinct real roots of the equation $x \sin x + \cos x = x^2$ is

- (A) 0 (B) 2 (C) 24 (D) none of the above.

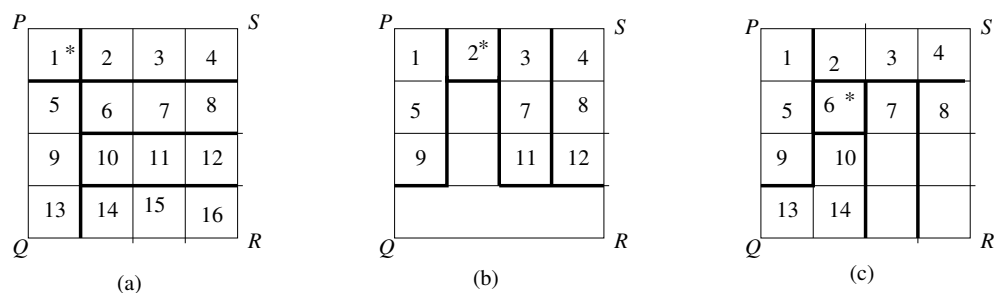
Answer and Comments: (B). By a direct substitution, 0 is not a root. Both the sides are even functions of x . So we might as well consider only positive roots, i.e. roots lying in the interval $(0, \infty)$. Let $f(x) = x^2 - x \sin x - \cos x$. Then $f(0) = -1 < 0$. Note that by the Cauchy Schwarz inequality, $|x \sin x + \cos x| \leq \sqrt{x^2 + 1}$. Hence $|x \sin x + \cos x| \leq 2x$ for all $x > 1$. Therefore $f(x) \rightarrow \infty$ as $x \rightarrow \infty$. So, by the Intermediate Value Property, $f(x)$ has at least one zero in $(0, \infty)$. But there cannot be more than 1 because $f'(x) = 2x - x \cos x > 0$ for all $x > 0$ and so f is strictly increasing on $(0, \infty)$.

Summing up, the given equation has exactly one positive root, say α . As said before, $-\alpha$ is also a root. Hence there are exactly two roots.

Q.18 You are given a 4×4 chessboard, and asked to fill it with five 3×1 pieces and one 1×1 piece. Then over all such fillings, the number of squares that can be occupied by the 1×1 piece is

- (A) 4 (B) 8 (C) 12 (D) 16.

Answer and Comments: (A). Number the sixteen squares of the 4×4 chessboard from 1 to 16 as shown in (a) of the figure below. These can be classified into three types : (A) the four corner squares (numbered 1, 4, 13 and 16), (B) the eight squares that lie along an edge but not at the corner (e.g. 2, 3, 5) and (C) the remaining four squares (numbered 6, 7, 10 and 11) that are in the interior. Any square of each type can be mapped onto any other square of the same type by a suitable symmetry of the chess-board. That helps to shorten the argument. We show the square occupied by the 1×1 piece by a putting a * inside it.



Surely, a filling is possible with the 1×1 piece occupying any of the four corners. By symmetry we take this square as numbered 1 and show the 3×1 pieces by thick boundaries as in (a) of the figure above.

But we claim that a filling is impossible with the 1×1 piece occupying a square of type (B) such as the square 2 in (b). Then the remaining squares in that row, viz. 1, 3 and 4 will have to be covered by 3×1 pieces placed vertically as shown in (b). Now there is no way to cover the remaining 6 squares by the two remaining 3×1 blocks.

Similarly, we claim that no filling is possible if the 1×1 piece occupies a square of type (C) such as 6 as in (c) above. Now we have a choice to cover square 1, either by a piece occupying 1, 5, 9 or by a

piece occupying squares 1, 2 and 3. By symmetry about the diagonal PR , we may take the first possibility as shown in (c). But then three 3×1 pieces will have to be placed as shown by thick lines and then there is no way to cover the remaining three squares (viz. 10, 13 and 14) by a single 3×1 piece.

Summing up, the 1×1 piece can be placed at any of the four corner squares but nowhere else.

A good problem. Although the argument takes a long time to write down, it does not take so long to conceive it. That makes the question ideal for an MCQ test.

- Q.19 A brand called Jogger's Pride produces pairs of shoes in three different units that are names U_1 , U_2 and U_3 . These units produce 10%, 30%, 60% of the total output of the brand with the chance that a pair of shoes being defective is 20%, 40%, 10% respectively. If a randomly selected pair of shoes from the combined output is found to be defective, then what is the chance that the pair was manufactured in the unit U_3 ?

- (A) 30% (B) 15%
 (C) $\frac{3}{5} \times 100\%$ (D) cannot be determined from the given data.

Answer and Comments: (A). Because of the uncertainties prevailing in all aspects of life, the probability problems are easily designed with a real life setting. However superficial the garb be, it provides a welcome relief from the drab world of ordered triples and existence of solutions. In the present problem, though, the paper setters have over stretched their license to choose a narrative of their liking. In a test where time is severely limited, it makes little sense to give the name of a brand. Also it is not clear what is gained by giving option (C) as $\frac{3}{5} \times 100\%$ instead of simply 60%.

Fortunately, the utter simplicity of the problem is an alleviating factor. It is a standard problem on conditional probability. Let U_1, U_2, U_3 also denote the events that a random pair of shoes comes from the respective units. We are given that

$$P(U_1) = \frac{1}{10}, P(U_2) = \frac{3}{10} \text{ and } P(U_3) = \frac{6}{10} \quad (1)$$

Let D be the event that the pair is defective. We are given the conditional probabilities

$$P(D|U_1) = \frac{2}{10}, P(D|U_2) = \frac{4}{10}, \text{ and } P(D|U_3) = \frac{1}{10} \quad (2)$$

The problem asks for the conditional probability $P(U_3|D)$, i.e. the probability that a pair comes from U_3 , given that it is defective. By Bayes theorem, this equals $\frac{P(U_3 \cap D)}{P(D)}$.

To find $P(D)$ we use that U_1, U_2, U_3 are mutually exclusive and exhaustive. Hence

$$P(D) = P(D \cap U_1) + P(D \cap U_2) + P(D \cap U_3) \quad (3)$$

The terms on the R.H.S. can be found from (1) and (2), using Bayes theorem again. Thus

$$P(D \cap U_1) = P(D|U_1)P(U_1) = \frac{2}{10} \times \frac{1}{10} = \frac{2}{100} \quad (4)$$

$$P(D \cap U_2) = P(D|U_2)P(U_2) = \frac{4}{10} \times \frac{3}{10} = \frac{12}{100} \quad (5)$$

$$\text{and } P(D \cap U_3) = P(D|U_3)P(U_3) = \frac{1}{10} \times \frac{6}{10} = \frac{6}{100} \quad (6)$$

Hence

$$P(D) = \frac{2}{100} + \frac{12}{100} + \frac{6}{100} = \frac{20}{100} \quad (7)$$

From (6) and (7),

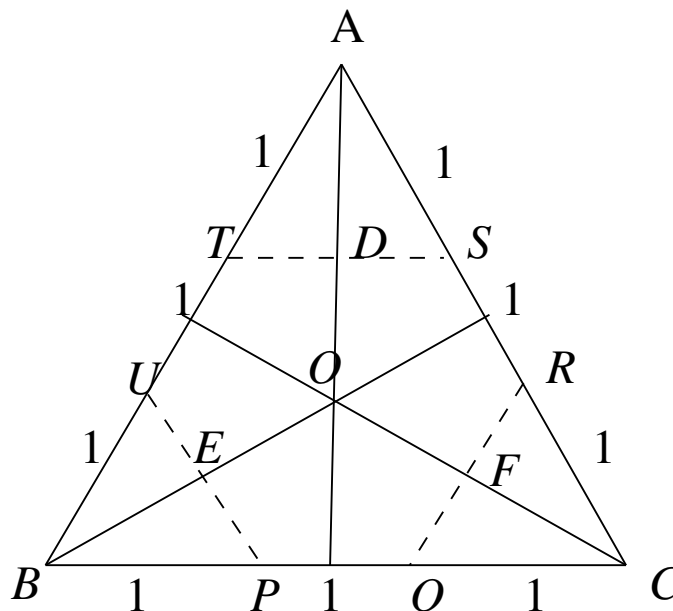
$$P(U_3|D) = \frac{P(U_3 \cap D)}{P(D)} = \frac{6/100}{20/100} = \frac{3}{10} = 30\% \quad (8)$$

A straightforward problem on conditional probability.

Q.20 Consider a paper in the shape of an equilateral triangle ABC with circumcentre O and perimeter 9 units. If we fold the paper in such a way that each of the vertices A, B, C gets identified with O , then the area of the resulting shape in square units is

- (A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{4}{\sqrt{3}}$ (C) $\frac{3\sqrt{3}}{2}$ (D) $3\sqrt{3}$.

Answer and Comments: (C). The first task is to identify the figure that results after folding. In the figure below, we show the original triangle ABC .



If the point A is to coincide with O after folding, the folding must take place along the perpendicular bisector of the segment AO , which is shown by the dotted line TS . It meets AO at its midpoint D . Similarly, the other two foldings must take place along the dotted lines UP and QR respectively, which pass through the midpoints E and F of BO and CO respectively.

The resulting figure after all three foldings is the hexagon $PQRSTU$. We claim that it is a regular hexagon with side 1. Clearly, all its angles are 120 degrees each. By symmetry, to show all sides are 1 unit, it suffices to show that two adjacent sides, say QR and RS , are of length 1 each.

As ABC is an equilateral triangle with side 3, each of its medians has length $\frac{\sqrt{3}}{2} \cdot 3$. Hence OC equals $\frac{2}{3} \times \frac{\sqrt{3}}{2} \times 3 = \sqrt{3}$. Hence $FC = \frac{\sqrt{3}}{2}$.

Therefore $QR = 2FR = 2FC \tan 30^\circ = \sqrt{3} \times \frac{1}{\sqrt{3}} = 1$. Also $RC = FC \sec 30^\circ = \frac{\sqrt{3}}{2} \times \frac{2}{\sqrt{3}} = 1$. By symmetry, $AS = CR = 1$. Hence $RS = 3 - 1 - 1 = 1$ unit.

The problem is now reduced to finding the area, of a regular hexagon with side 1. This is 6 times the area of an equilateral triangle with side 1. It comes out to be $6 \times \frac{1}{2} \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$.

A good problem which tests the ability to visualise the figure that results after the folding. The numerical data has been carefully chosen to avoid messy calculations.

- Q.21 Let P be a regular twelve-sided polygon. The number of right angled triangles formed by the vertices of P is

(A) 60 (B) 120 (C) 100 (D) 220.

Answer and Comments: (A). The vertices of P all lie on some circle, say C . Three such vertices will form a right angled triangle if and only if two of them are diametrically opposite. There are in all 6 diameters, each containing two vertices of P . For each such diameter, the third vertex can be any of the remaining 10 vertices of P . Hence the total number of triangles is $6 \times 10 = 60$.

A simple problem once the key idea that the angle in a semi-circle is a right angle strikes.

- Q.22 If the n terms $a_1, a_2, \dots, a_n$ are in arithmetic progression with increment r , then the difference between the mean of their squares and the square of their mean is

(A) $\frac{r^2((n-1)^2 - 1)}{12}$ (B) $\frac{r^2}{12}$
 (C) $\frac{r^2(n^2 - 1)}{12}$ (D) $\frac{n^2 - 1}{12}$

Answer and Comments: (C). Take the terms of the A.P. as $a, a + r, a + 2r, \dots, a + (n-1)r$. Then their mean, say m , is

$$m = a + \frac{(n-1)r}{2} \tag{1}$$

Hence

$$m^2 = a^2 + ar(n-1) + \frac{(n-1)^2 r^2}{4} \quad (2)$$

The squares of the terms of the A.P. are $a^2, (a+r)^2, \dots, (a+(n-1)r)^2$. Hence their mean, say M is

$$\begin{aligned} M &= \frac{1}{n} \sum_{k=0}^{n-1} (a + kr)^2 \\ &= \frac{1}{n} \sum_{k=0}^{n-1} (a^2 + 2akr + k^2 r^2) \\ &= \frac{1}{n} \left[a^2 n + 2ar \frac{n(n-1)}{2} + r^2 \frac{(n-1)n(2n-1)}{6} \right] \\ &= a^2 + ar(n-1) + r^2 u \end{aligned} \quad (3)$$

From (2) and (3),

$$\begin{aligned} M - m^2 &= r^2 \left[\frac{(n-1)(2n-1)}{6} - \frac{(n-1)^2}{4} \right] \\ &= \frac{r^2(n-1)}{12} (4n-2-3n+3) \\ &= \frac{r^2(n^2-1)}{12} \end{aligned} \quad (4)$$

which tallies with (A).

An absolutely computational problem requiring little more than the definition of an A.P. and the formulas for the sums $\sum_{k=0}^{n-1} k$ and

$$\sum_{k=0}^{n-1} k^2.$$

- Q.23 A father wants to distribute a certain sum of money between his daughter and son in such a way that if both of them invest their shares in the scheme that offers compound interest at $\frac{25}{3}\%$ per annum, for t and $t+2$ years respectively, then the two shares grow to become equal. If the son's share was rupees 4,320, then the total money distributed by the father was

- (A) rupees 9360 (B) rupees 9390
 (C) rupees 16590 (D) rupees 16640.

Answer and Comments: (B). It is not given but it is to be assumed that the interest is compounded annually. At the rate of $\frac{25}{3}\%$ per annum, x rupees give a total of $x + \frac{25}{300}x = \frac{13}{12}x$ rupees at the end of the year. So after $t + 2$ years, the son's share becomes $4320 \times (\frac{13}{12})^{t+2}$ rupees. If we denote the daughter's share by y , at the end of t years, it will become $y(\frac{13}{12})^t$ rupees. This gives an equation

$$y(\frac{13}{12})^t = 4320(\frac{13}{12})^{t+2} \quad (1)$$

Canceling $(\frac{13}{12})^t$ from both the sides this becomes

$$\begin{aligned} y &= 4320(\frac{13}{12})^2 = 4320 \times \frac{169}{144} \\ &= \frac{360}{12} \times 169 \\ &= 30 \times 169 = 5070 \end{aligned} \quad (2)$$

Hence the total money distributed by the father is $5070 + 4320 = 9390$ rupees.

Essentially, a high school level problem. The only catch is to recognise that the value of t is not needed.

Q.24 Let α be a real number. The range of values of $|\alpha - 4|$ such that $|\alpha - 1| + |\alpha + 3| \leq 8$ is

- (A) $(0, 7)$ (B) $(1, 8)$ (C) $[1, 9]$ (D) $[2, 5]$.

Answer and Comments: (C). Think of $|\alpha - 4|$ as a function, say $f(\alpha)$ of a real variable α . Let D be the set $\{\alpha \in \mathbb{R} : |\alpha - 1| + |\alpha + 3| \leq 8\}$. The problems asks to find the set $f(D)$, i.e. the image of D under the function D .

We begin by expressing D in a simplified form. Call the expression $|\alpha - 1| + |\alpha + 3|$ as $g(\alpha)$. Then it changes its formula at $\alpha = -3$ and again at $\alpha = 1$. So we study it separately for the three intervals

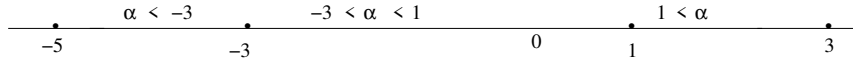
$\alpha \leq -3$, $-3 \leq \alpha \leq 1$ and $\alpha \geq 1$ to find their intersections with the set D .

For $\alpha \leq -3$, $\alpha \leq 1$ also holds and so we have $g(\alpha) = 1 - \alpha - 3 - \alpha = -2 - 2\alpha$. So $g(\alpha) \leq 8$ will be satisfied if and only if $-2 - 2\alpha \leq 8$, i.e. $\alpha \geq -5$. This gives us the interval $[-5, -3]$ as a part of the domain D .

For $-3 < \alpha < 1$, $g(\alpha) = \alpha + 3 + 1 - \alpha = 2$ which is a constant. So the condition $g(\alpha) \leq 8$ holds for all points in the interval and hence they all lie in D .

Finally, for $\alpha \geq 1$, $\alpha \geq -3$ also holds and so $g(\alpha) = \alpha - 1 + \alpha + 3 = 2\alpha + 2$. So $g(\alpha) \leq 8$ holds for $\alpha \leq 3$.

Summing up, $D = [-5, -3] \cup (-3, 1) \cup [1, 3] = [-5, 3]$.



It is now easy to finish the solution. The function $f(\alpha) = |\alpha - 4|$ changes its formula at $\alpha = 4$. But the point 4 is outside and to the right of D . So, for all $\alpha \in D$, we have $f(\alpha) = 4 - \alpha$. This is a strictly monotonically decreasing function with $f(-5) = 9$ and $f(3) = 1$. So, $f(D)$ is the interval $[1, 9]$.

A straightforward problem but requiring rather repetitious reasoning. It would have been a little more interesting if instead of $|\alpha - 4|$, $f(\alpha)$ were given as say, $|3\alpha - 4|$, because then the point $\frac{4}{3}$ at which f changes its formula would lie inside D and so separate considerations would be needed for the two parts of D , one to the left of $\frac{4}{3}$ and the other to its right. The first part would be mapped onto $[0, 19]$ and the second to $[0, 5]$. So the answer would be the interval $[0, 19]$.

There is a novel way (suggested by Deepanshu Rajvanshi) to identify the set $D = \{\alpha \in \mathbb{R} : |\alpha - 1| + |\alpha + 3| \leq 8\}$ if we replace α by a complex number z . Then $D = \mathbb{R} \cap E$, where $E = \{z \in \mathbb{C} : |z - 1| + |z + 3| \leq 8\}$. In general, identifying E would be more complicated than identifying D . But in the present problem, E has a nice geometric interpretation if we recognise $|z - 1|$ and $|z + 3|$ as the distances of z from the fixed points 1 and -3 in the complex plane. Then E is the inside of the ellipse, say F with foci at 1 and -3 and the major axis $2a = 8$, so that $a = 4$. The centre of the ellipse is at -1 (at the middle of the two foci).

Hence the major axis of the ellipse is from $-1 - 4$ to $-1 + 4$, i.e. from -5 to 3 . So, $D = [-5, 3]$.

- Q.25 For each natural number k , choose a complex number z_k with $|z_k| = 1$ and denote by a_k , the area of the triangle formed by z_k, iz_k and $z_k + iz_k$. Then which of the following is true for the series below ?

$$\sum_{k=1}^{\infty} (a_k)^k$$

- (A) It converges only if every z_k lies in the same quadrant.
- (B) It always diverges.
- (C) It always converges.
- (D) none of the above.

Answer and Comments: (C). A masterpiece of a deliberately misleading formulation. For any non-zero complex number z , the points $z, 0, iz$ and $z + iz$ form a square. This follows because if P, Q are the points represented by z and iz respectively in the Argand diagram, then the vector $\vec{OQ}$ is obtained from the vector $\vec{OP}$ by counterclockwise rotation around O through 90 degrees. As a result, the area of the triangle with vertices z, iz and $z + iz$ is simply $\frac{1}{2}|z|^2$.

In the present problem $|z_k| = 1$ for all k . Hence $a_k = \frac{1}{2}$ and the series $\sum_{k=1}^{\infty} (a_k)^k$ is a geometric series with common ratio $\frac{1}{2}$. Obviously it is convergent.

The problem has very little to do with complex numbers, or with geometry. It involves infinite series in a very shallow way.

- Q.26 The function $y = e^{kx}$ satisfies the differential equation

$$\left(\frac{d^2y}{dx^2} + \frac{dy}{dx}\right)\left(\frac{dy}{dx} - y\right) = y\frac{dy}{dx}$$

for

- (A) exactly one value of k .
- (B) two distinct values of k .
- (C) three distinct values of k .
- (D) infinitely many values of k .

Answer and Comments: (C). By a direct calculation, $\frac{dy}{dx} = ke^{kx} = ky$ and $\frac{d^2y}{dx^2} = k^2e^{kx} = k^2y$. Putting these into the given differential equation, we get

$$(k^2 + k)(k - 1)y^2 = y^2k \quad (1)$$

As the exponential function never vanishes, we get $k(k^2 - 1) = k$, which has three distinct roots, viz. $0, \sqrt{2}$ and $-\sqrt{2}$. Each of these gives a solution. (For $k = 0$, y is the constant function 1. Still it is a solution since both the sides of the equation are 0.)

A trivial problem.

Q.27 For a real number θ consider the following simultaneous equations:

$$\begin{aligned} (\cos \theta)x - (\sin \theta)y &= 1 \\ (\sin \theta)x + \cos \theta)y &= 2 \end{aligned}$$

The number of solutions of these equations in x and y is

- (A) 0
- (B) 1
- (C) infinite for some values of θ
- (D) finite only when $\theta = \frac{m\pi}{n}$ for integers m and $n \neq 0$.

Answer and Comments: (B). The coefficient matrix is $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$.

Its determinant is $\cos^2 \theta + \sin^2 \theta$ which is always non-zero (in fact equals 1). Hence there is a unique solution.

Another absolutely simple problem. One fails to see the purpose of asking such questions.

Q.28 In the range $0 \leq x \leq 2\pi$, the equation $\cos(\sin x) = \frac{1}{2}$ has

- (A) 0 solutions.
- (B) 2 solutions.

- (C) 4 solutions.
 (D) infinitely many solutions.

Answer and Comments: (A). Since $\frac{1}{2} = \cos(\frac{\pi}{3})$, $\cos(\sin x)$ equals $\frac{1}{2}$ if and only if $\sin x = 2n\pi \pm \frac{\pi}{3}$ for some integer n . But no matter what n is, $2n\pi \pm \frac{\pi}{3}$ lies outside the interval $[-1, 1]$ which is the range of the sine function. (Here we are using that $3 < \pi < 4$.) So there is no value of x which satisfies $\sin x = 2n\pi \pm \frac{\pi}{3}$ for any integer n . So the given equation has no solution.

Yet another trivial problem.

Q.29 A particle is allowed to move in the XY -plane by choosing any one of the two jumps:

1. Move two units to the right and one unit up, i.e. $(a, b) \mapsto (a + 2, b + 1)$, or
2. Move two units up and one unit to the right, i.e. $(a, b) \mapsto (a + 1, b + 2)$.

Let $P = (30, 63)$ and $Q = (100, 100)$. If the particle starts at the origin, then

- (A) P is reachable but not Q .
 (B) Q is reachable but not P .
 (C) both P and Q are reachable.
 (D) neither P nor Q is reachable.

Answer and Comments: (D). Suppose a point (x, y) is reached by taking m jumps of the first type and n jumps of the second type. Then x is the total displacement to the right and y is the total displacement upwards. This gives a system of equations

$$2m + n = x \tag{1}$$

$$\text{and } m + 2n = y \tag{2}$$

This is a system of two linear equations in the unknowns m and n . It can be solved by inspection to give unique solutions for m and n , viz.

$$m = \frac{2x - y}{3} \quad (3)$$

$$\text{and } n = \frac{x - 2y}{3} \quad (4)$$

When $(x, y) = (30, 63)$, the solution is $m = -1$ and $n = 32$. But m, n have to be non-negative integers, because a reverse jump is not allowed. So the point $(30, 63)$ is not reachable.

When $(x, y) = (100, 100)$, the solution is $m = \frac{100}{3}, n = \frac{100}{3}$. Since neither is an integer, the point Q is not reachable either.

An absolutely simple problem. A far more challenging problem of a similar spirit was asked in the CMI Entrance Test in 2016. It goes as follows.

“ A *step* starting at a point P in the XY -plane consists of moving by one unit from P in one of three directions: directly to the right or in the direction of one of the two rays that make the angle of 120 degrees with positive X-axis. (An opposite move, i.e. to the left/southeast/northeast, is not allowed.) A *path* consists of a number of such steps, each new step starting where the previous step ended. Points and steps in a path may repeat. Find the number of paths starting at $(1, 0)$ and ending at $(2, 0)$ that consist of (i) exactly 6 steps (ii) exactly 7 steps.”

The answers to (i) and (ii) are 0 and 210 respectively.

Q.30 For a real polynomial in one variable P , let $Z(P)$ denote the locus of points (x, y) in the plane such that $P(x) + P(y) = 0$. Then,

- (A) there exist polynomials Q_1 and Q_2 such that $Z(Q_1)$ is a circle and $Z(Q_2)$ is a parabola.
- (B) There does not exist any polynomial Q such that $Z(Q)$ is a circle or a parabola.
- (C) there exists a polynomial Q such that $Z(Q)$ is a circle but there does not exist any polynomial P such that $Z(P)$ is a parabola.

- (D) there exists a polynomial Q such that $Z(Q)$ is a parabola but there does not exist any polynomial P such that $Z(P)$ is a circle.

Answer and Comments: (C). Another example where a question whose underlying idea is very simple is given a formidably verbose formulation. The general form of the equation of a circle is $x^2 + y^2 + 2gx + 2fy + c = 0$. The question is whether with a suitable choice of the constants f, g and c , we can write this in the form $Q(x) + Q(y) = 0$ for a suitable polynomial $Q(x)$. Obviously, this is possible by taking $f = g = 0$ and c to be -1 . (Any negative number will do.) Take $Q(x) = x^2 - \frac{1}{2}$. Then $Q(x) + Q(y) = 0$ represents the unit circle $x^2 + y^2 = 1$.

As for a parabola, the general equation is $Ax^2 + 2Hxy + By^2 + 2gx + 2fy + c = 0$ where $H^2 = AB$. If $H = 0$, then no matter which $P(x)$ we take, the expression $P(x) + P(y)$ cannot equal $Ax^2 + 2Hxy + By^2 + 2gx + 2fy + c = 0$. On the other hand, suppose $H \neq 0$. Then $AB \neq 0$ and so at least one of A and B is non-zero. But both cannot be zero as otherwise the parabola degenerates into a straight line. Suppose, without loss of generality, that $A \neq 0$ and $B = 0$. But then in $P(x)$ the coefficient of x^2 will be A and so in $P(x) + P(y)$, the coefficient of y^2 would be A too, a contradiction.

Another absolutely trivial problem. Probably the purpose of asking it is to test whether the candidates can understand mathematical statements and distinguish those who do from those who can do only computational problems.

ISI BStat-BMath-UGA-2019 Paper 2 with Comments

Q.1 Prove that the positive integers n that cannot be written as a sum of r consecutive positive integers, with $r > 1$, are of the form $n = 2^l$ for some $l \geq 0$.

Solution and Comments: We prove the contrapositive. That is, suppose n is not of the form 2^l for any $l \geq 0$. Then n has at least one odd factor $r > 1$ dividing it. Let $r = 2k + 1$. Let $m = \frac{n}{r}$. Then n is a sum of r consecutive integers with the middle integer being m . That is, $n = (m - k) + \dots + (m - 1) + m + (m + 1) + \dots + (m + k)$. If $m > k$, then this is a sum of $2m + 1$ consecutive positive integers which adds to n . Even if $m \leq k$, so that some of the initial terms are negative or 0, the negative ones will get cancelled with the positive ones and the remaining consecutive positive integers will add to n . (For example, if $n = 14$, then $r = 7, k = 3$ and $m = 2$. So $14 = -1 + 0 + 1 + 2 + 3 + 4 + 5 = 2 + 3 + 4 + 5$.) ■

Although not asked, the converse is also true. That is, an integer of the form 2^l cannot be expressed as a sum of consecutive positive integers. For suppose $n = (s + 1) + (s + 2) + \dots + (t - 1) + t$ (say) where $0 \leq s < t - 1$, then $n = \frac{t(t + 1) - s(s + 1)}{2}$. Hence $2n = (t - s)(t + s + 1)$. Out of these two factors, one is odd and the other even. So $2n$ and hence n has at least one odd factor greater than 1.

An easy problem, as the answer is given. A better problem would have been to characterise all positive integers which can be expressed as a sum of more than one consecutive positive integers.

Q.2 Let $f : (0, \infty) \rightarrow \mathbb{R}$ be defined by

$$f(x) = \lim_{n \rightarrow \infty} \cos^n\left(\frac{1}{n^x}\right).$$

- (a) Show that f has exactly one point of discontinuity.
- (b) Evaluate f at its point of discontinuity.

Solution and Comments: The very definition of $f(x)$ is by a limit which is in the 1^∞ form. So we evaluate it using logarithms. Thus

$$f(x) = e^{g(x)} \quad (1)$$

where

$$g(x) = \lim_{n \rightarrow \infty} n \ln(\cos(\frac{1}{n^x})) \quad (2)$$

for $x > 0$. If we let $u = \frac{1}{n}$ then $u \rightarrow 0^+$ as $n \rightarrow \infty$. So, if we can compute $h(x)$ where

$$h(x) = \lim_{u \rightarrow 0^+} \frac{\ln(\cos(u^x))}{u} \quad (3)$$

then surely $h(x)$ will equal $g(x)$ because in (3) $u \rightarrow 0^+$ through all real values and not just those of the form $\frac{1}{n}$. (Such fine points are usually ignored when one indeterminate form is converted to another, because they do not affect the limit.)

Applying l'Hôpital's rule,

$$h(x) = \lim_{u \rightarrow 0^+} -\tan(u^x) x u^{x-1} \quad (4)$$

Near 0, $\tan(u^x)$ is comparable to u^x for $u > 0$ (i.e. their ratio tends to a finite non-zero limit, viz. 1). So, we get

$$h(x) = - \lim_{u \rightarrow 0^+} x u^{2x-1} \quad (5)$$

We now make three cases depending upon where x lies.

- (i) for $x > \frac{1}{2}$, the exponent $2x - 1$ is positive and so $u^{2x-1} \rightarrow 0$ as $y \rightarrow 0^+$. Hence $h(x) = 0$.
- (ii) for $x = \frac{1}{2}$, $u^0 = 1$ and so $h(x) = -\frac{1}{2}$.
- (iii) for $0 < x < \frac{1}{2}$, the exponent $2x - 1$ is negative and so $u^{2x-1} \rightarrow \infty$ as $u \rightarrow 0^+$.

Summing up, and applying (1)

$$f(x) = e^{g(x)} = e^{h(x)} = \begin{cases} 0, & 0 < x < \frac{1}{2} \\ e^{-1/2}, & x = \frac{1}{2} \\ 1, & x > \frac{1}{2} \end{cases} \quad (6)$$

It is now obvious that f only one point of discontinuity, viz. $\frac{1}{2}$ and the value of f at this point is $e^{-1/2}$. ■

A good problem on limits in indeterminate forms.

Q.3 Let $\Omega = \{z = x + iy \in \mathbb{C} : |y| \leq 1\}$. If $f(z) = z^2 + 2$, then draw a sketch of

$$f(\Omega) = \{f(z) : z \in \Omega\}.$$

Justify your answer.

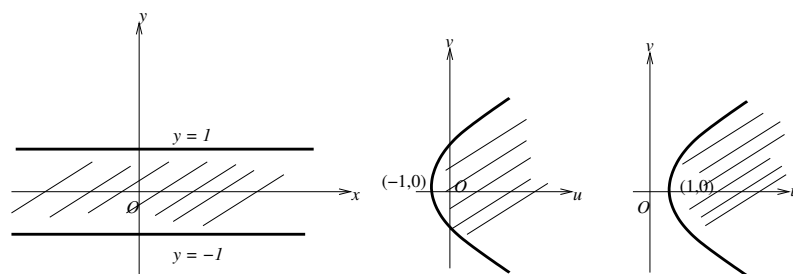
Solution and Comments: Let $g(z) = z^2$. Then $f(z) = g(z) + 2$. So the real task is to identify $g(\Omega)$. Once that is done, merely shift every point to the right by 2 units.

To find $g(\Omega)$, we let $g(z) = u + iv$. Then

$$u = x^2 - y^2 \quad (1)$$

$$\text{and } v = 2xy \quad (2)$$

The set Ω is an infinite horizontal strip $-1 \leq y \leq 1$ shown in (a) of the figure below. Its upper boundary, say L_1 , is the line $y = 1$.



(a) Ω

(b) $g(\Omega)$

(c) $f(\Omega)$

Under g , L_1 is mapped to the curve in the uv -plane whose parametric equations are

$$u = x^2 - 1, v = 2x, \quad -\infty < x < \infty \quad (3)$$

Eliminating x , we get

$$4u = v^2 - 4 \quad (4)$$

which is a parabola, $v^2 = 4(u+1)$, centred at $(-1, 0)$ and focus at the origin. It is shown by a thick curve in (b). Call it C_1 .

Similarly, the lower boundary of Ω is the line $y = -1$. Under g it is mapped to a curve, say C_2 , whose parametric equations are

$$u = x^2 - 1, \quad v = -2x \quad (5)$$

Eliminating x , this becomes $v^2 = 4(u+1)$. So this is the same parabola as C_1 . This may seem baffling, but is not so if we keep in mind that $g(-z) = g(z)$. If $z \in L_1$ then $-z \in L_2$ and so it is not surprising that g maps both the lines L_1 and L_2 to the same parabola C_1 .

More generally, if z is in the upper half of Ω , then $-z$ is in the lower half and does not contribute anything new to $g(\Omega)$. So to know $g(\Omega)$ and hence $f(\Omega)$, it suffices to consider only the images of the lines of the form $y = c$ for $0 \leq c \leq 1$. They are all parabolas centred at $(-\frac{c^2}{2}, 0)$ for $0 < c \leq 1$. The image of $y = 0$ (i.e. the x -axis) under g is the non-negative u -axis.

$g(\Omega)$ is shown in (b) of the figure. As noted earlier, $f(\Omega)$ is obtained from $g(\Omega)$ by a mere shift of 2 units to the right. It is a region bounded by the parabola $v^2 = 4(u-1)$. It is shown in (c).

Q.4 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that

$$\frac{1}{2y} \int_{x-y}^{x+y} f(t) dt = f(x), \text{ for all } x \in \mathbb{R}, y > 0.$$

Show that there exist $a, b \in \mathbb{R}$ such that $f(x) = ax + b$ for all $x \in \mathbb{R}$.

Solution and Comments: Write the given condition as

$$\int_{x-y}^{x+y} f(t) dt = 2yf(x) \quad (1)$$

for all $y > 0$. Since $\int_{x+y}^{x-y} f(t) dt = -\int_{x-y}^{x+y} f(t) dt$, we see that (1) is valid for $y < 0$ too and hence for all $y \in \mathbb{R}$.

We now fix any x and differentiate both the sides of (1) w.r.t. y using the fundamental theorem of calculus (in Leibnitz form where both the upper and the lower limits of the integral are variables). Then,

$$f(x+y) + f(x-y) = 2f(x) \quad (2)$$

for all x, y . Differentiating w.r.t. y again,

$$f'(x+y) - f'(x-y) = 0 \quad (3)$$

for all x, y . This means that f' is constant because given any a, b with $a < b$, we can write $a = x-y$ and $b = x+y$ where $x = \frac{a+b}{2}$ and $y = \frac{b-a}{2}$. But if $f'(x)$ is a constant, then by integrating it we see that $f(x) = ax + b$ for some constants a, b . ■

The fact that f'' exists was never used in the proof above. For applying the Fundamental Theorem of Calculus, we need that f is continuous. Then in deriving (3) from (2), we used that f is differentiable. Actually, even that is not necessary. It is possible to get the result directly from (2) using it repeatedly. Verbally, (2) says that the value of f at the A.M. of any two numbers, say x and y , equals the A.M. of its values at those two numbers. Applying this twice we can show that for any $x < y$,

$$f\left(\frac{1}{4}x + \frac{3}{4}y\right) = \frac{1}{4}f(x) + \frac{3}{4}f(y) \quad (4)$$

All we need is to notice is that $\frac{x+3y}{4}$ is the A.M. of $\frac{x+y}{2}$ and y . Similarly,

$$f\left(\frac{3}{4}x + \frac{1}{4}y\right) = \frac{3}{4}f(x) + \frac{1}{4}f(y) \quad (5)$$

Repeated applications give that for any dyadic rational $r \in (0, 1)$ (i.e. a rational number whose denominator in its reduced form is a power of 2),

$$f(rx + (1-r)y) = rf(x) + (1-r)f(y) \quad (6)$$

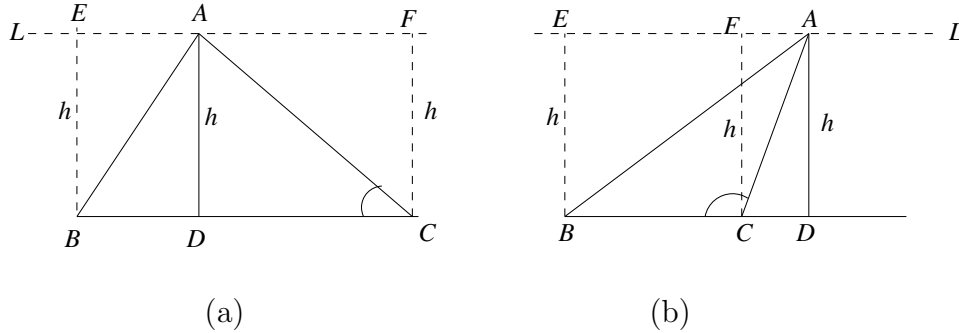
Since the set of dyadic rational numbers in $(0, 1)$ is dense in $[0, 1]$, continuity of f implies that (6) holds true for any $r \in [0, 1]$. But then the graph of f is a straight line. Hence $f(x)$ is of the form $ax + b$ for some constants a, b .

Q.5 A subset S of the plane is called *convex* if given any two points x and y in S , the line segment joining x and y is contained in S . A quadrilateral is called convex if the region enclosed by the edges of the quadrilateral is a convex set.

Show that given a convex quadrilateral Q of area 1, there is a rectangle R of area 2 such that Q can be drawn inside R .

Solution and Comments: The piece of data that the quadrilateral Q has area 1 is not particularly relevant. We shall ignore it and show that if Q is any convex quadrilateral, then there is a rectangle R containing it whose area is twice that of Q .

The argument uses an extension of the construction in deriving the formula for the area of a triangle from that of a rectangle. Suppose ABC is a triangle. Through A draw a line L parallel to the base BC . Drop perpendiculars BE and CF from B, C to L and a perpendicular AD from A to BC as shown in (a) of the figure below. Then $\triangle ABC$ is contained in the rectangle $BCFE$ whose area is $h \cdot BC$ where $h = AD = BE = CF$. And we get the area of the triangle ABC by adding the areas of the triangles ABD and ADC , each of which is half the area of the rectangle $ADBE$ and $ADCF$ respectively. Thus $BCFE$ is a rectangle containing the triangle ABC and area twice that of the triangle ABC .



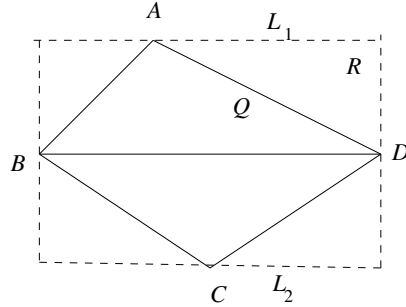
But this argument is valid only when both the base angles B and C are acute. If one of them, say C , is obtuse as shown in (b) above, then the perpendicular AD falls on the extended side BC . We still get the area of the triangle ABC by subtracting the area of ACD from that of ABD . But now the rectangle $BCFE$ no longer contains the triangle ABC . The rectangle $BDAE$ does contain the triangle ABC , but its area exceeds twice the area of the triangle ABC .

We can avoid this difficulty by noting that a triangle can have at most one obtuse angle and supposing, without loss of generality, that

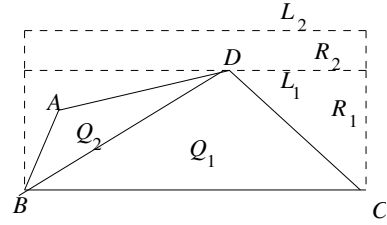
it is A . But when we try a similar construction for a given convex quadrilateral $Q = ABCD$ (say), all we know is that the four angles add to 360 degrees. Hence as many as three of them can be obtuse. Also there is at least one obtuse angle except when $ABCD$ is itself a rectangle. When this is the case, we can double it along any edge to get a rectangle containing it and having twice the area.

So we have to tackle the cases where $Q = ABCD$ has one, two or three obtuse angles. In each case we shall find a rectangle R containing Q and having double the area of Q , with one side parallel to some diagonal of Q or some edge of Q .

The simplest case is when only one angle, say A , is obtuse. In this case we modify the argument above for obtaining the area of a triangle. By convexity, the vertices A and C will lie on the opposite sides of the diagonal BD . Through A and C draw lines L_1 and L_2 respectively, parallel to BD as shown in (a) below. Also draw lines through B and D perpendicular to L_1, L_2 . Then the rectangle R bounded by these four lines is a desired rectangle.



(a)



(b)

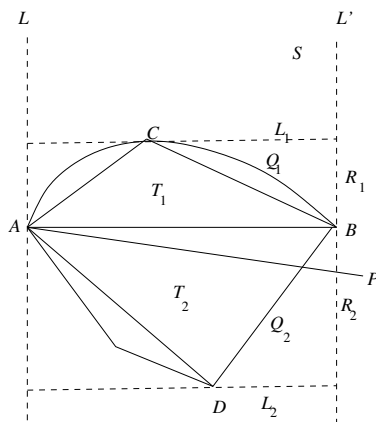
In fact, this construction will apply even when, in addition to A , the opposite angle C and possibly one of the remaining two angles is obtuse. For obtuseness of A and C ensures that the base angles of both the triangles ABD and CBD are acute. Note that in all these cases Q is not only contained in the resulting rectangle R , it is inscribed in R , i.e. all the vertices of Q lie on the sides of R .

The only case left is when two adjacent angles, say A and D , of Q are obtuse and the other two not obtuse. In this case, A and D lie

on the same side of BC . Without loss of generality, assume that D is farther from BC than A as shown in (b) of the figure above. Divide Q into two triangles Q_1 and Q_2 by the diagonal BD . Draw a line L_1 through D parallel to BC and drop perpendiculars from B and C onto this line. Then we get a rectangle R_1 whose area is twice that of Q_1 . Note that R_1 also contains Q_2 because $\angle ABC$ is at most 90 degrees. But the area of R_1 falls short of twice the area of Q . To correct this, with L_1 as a base, draw another rectangle R_2 (on the opposite side as R_1) whose area is twice that of Q_2 . Let $R = R_1 \cup R_2$. Then R is a desired rectangle. Note that this time although R contains Q , Q is not inscribed in R .

Thus we have proved the existence of a desired rectangle in all cases. Actually, all the cases considered follow as special cases of a more general result. Suppose Q is a bounded, closed convex set in the plane. We claim that Q is contained in a rectangle R whose area is at most twice that of Q . (This more general problem appears as Q.21 in *Problem Solving Strategies* by Arthur Engel, a Springer publication.) Here, ‘closed’ is a technical term, which loosely means a set which contains all its boundary points. For example, if you take a triangle T or a disc D with all their boundary points in, they are closed sets. But if you remove some of the boundary points, then they are no longer closed sets.

Now suppose Q is a bounded, closed convex set in the plane. Let A and B be two points of Q that are farthest apart from each other. (The existence of such a pair is intuitively obvious. A rigorous proof is not so trivial and uses what is called the **completeness** of the real number system, a basic property which is needed in the proof of many other results of theoretical calculus, such as the Intermediate Value Property.) Join AB and draw lines L and L' through A and B respectively that are perpendicular to AB as shown below.



First we claim that Q is contained in the strip, say S , bounded by these two parallel lines. For suppose some point P outside S is in Q . Then its distance from at least one of A and B exceeds the length of AB , contradicting the maximality of AB . Now let Q_1 and Q_2 be the portions of Q lying on the two sides of AB . (It is possible that one of these two subsets contains no points except those on AB .) Let C and D be the points of Q_1 and Q_2 that are farthest from the line AB . Draw lines L_1 and L_2 through C and D respectively that are parallel to AB . Let T_1 and T_2 be respectively, the triangles ABC and ABD . By convexity of Q , $T_1 \subset Q_1$ and $T_2 \subset Q_2$. Let R_1 be the rectangle formed by the lines AB, L, L' and L_1 and R_2 be the rectangle formed by the lines AB, L, L' and L_2 . Then $Q_1 \subset R_1$, as otherwise Q_1 will contain a point which is farther from AB than C is. Also the area of R_1 is twice that of T_1 . But $T_1 \subset Q_1$. Hence the area of R_1 is at most twice that of Q_1 . Similarly, the area of R_2 is at most twice that of Q_2 . Therefore $R = R_1 \cup R_2$ is a rectangle containing Q whose area is at most twice that of Q . (If we want, we can widen R vertically to get a rectangle whose area is exactly twice that of Q .)

The technique used in this argument is called the **extremality principle** because we began by choosing a farthest pair of points of Q . (Such a pair need not be unique, as happens when Q is a closed disc, for example. Also sometimes, an extremum can be a minimum.) Later in the proof, we chose C and D to be the points of Q_1, Q_2 respectively whose distances from the line AB are maximum possible. (Again such points may not be unique.)

Q.6 For all natural numbers n , let

$$A_n = \sqrt{2 - \sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}} \quad (n \text{ many radicals}).$$

(a) Show that for $n \geq 2$,

$$A_n = 2 \sin \frac{\pi}{2^{n+1}}.$$

(b) Hence, or otherwise, evaluate the limit

$$\lim_{n \rightarrow \infty} 2^n A_n.$$

Solution and Comments: We note that inside the first radical sign, the radical appears with a negative sign. But all subsequent radicals appear with a plus sign. So it is convenient to introduce another number defined through n radicals. Specifically, we let

$$B_n = \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}} \quad (n \text{ many radicals}). \quad (1)$$

Then $B_1 = \sqrt{2}$, $B_2 = \sqrt{2 + \sqrt{2}}$ and so on.

Note that

$$A_n^2 = 2 - B_{n-1} \quad (2)$$

So, proving (a) is equivalent to showing that

$$2 - B_{n-1} = 4 \sin^2 \frac{\pi}{2^{n+1}} = 2(1 - \cos \frac{\pi}{2^n}) \quad (3)$$

which, in turn, is equivalent to showing that

$$B_n = 2 \cos \frac{\pi}{2^{n+1}} \quad (4)$$

for all positive integers n . This can be done by induction on n . We already have $B_1 = \sqrt{2} = 2 \cos \frac{\pi}{4}$. So the statement is true for $n = 1$.

For the inductive step, we merely note that

$$\begin{aligned}
 B_{k+1} &= \sqrt{2 + B_k} \\
 &= \sqrt{2 + 2 \cos \frac{\pi}{2^{k+1}}} \\
 &= \sqrt{2 \times 2 \cos^2 \frac{\pi}{2^{k+2}}} \\
 &= 2 \cos \frac{\pi}{2^{k+2}}
 \end{aligned} \tag{5}$$

Thus we have proved (4) for all n , and as noted already it completes the proof of (a).

For (b) we use that $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$. By (a),

$$\begin{aligned}
 2^n A_n &= 2^{n+1} \sin \frac{\pi}{2^{n+1}} \\
 &= \pi \times \frac{\sin(\pi/2^{n+1})}{\pi/2^{n+1}}
 \end{aligned} \tag{6}$$

As $n \rightarrow \infty$, $\pi/2^{n+1} \rightarrow 0$ and so the second factor tends to 1. Therefore the limit in (b) is π . ■

Q.7 Let f be a polynomial with integer coefficients. Define

$$a_1 = f(0), a_2 = f(a_1) = f(f(0)),$$

and

$$a_n = f(a_{n-1}) \text{ for } n \geq 3.$$

If there exists a natural number $k \geq 3$ such that $a_k = 0$, then prove that either $a_1 = 0$ or $a_2 = 0$.

Solution and Comments: For notational uniformity, let $a_0 = 0$. Then $a_n = f(a_{n-1})$ for all $n \geq 1$. Let $d_n = a_n - a_{n-1}$ for $n \geq 1$. We note that for any polynomial $p(x)$ and for any two real numbers a and b , $(b - a)$ is a factor $p(b) - p(a)$. Moreover, the ratio is an integer if both a, b and the coefficients in $p(x)$ are all integers.

Since

$$d_{n+1} = a_{n+1} - a_n = f(a_n) - f(a_{n-1}) \tag{1}$$

and $d_n = a_n - a_{n-1}$ we get that d_n divides d_{n+1} as an integer (because f is a polynomial with integer coefficients.)

Now, suppose $a_k = 0$ for some $k \geq 3$. Then $a_k = a_0$ and hence $a_{k+1} = a_1$, $a_{k+2} = a_2$ and so on. So the numbers $a_1, a_2, \dots, a_n$ and hence also the differences $d_1, d_2, \dots, d_n$ recur in a cycle of length k . But that means d_k divides d_1 since $d_1 = d_{k+1}$. Thus in the sequence $d_1, d_2, d_3, \dots, d_k, d_1$ every integer divides the next one. If one of them is 0, so are all others. When this happens, $a_1 = d_1 = 0$. Otherwise let $r_i = \frac{d_{i+1}}{d_i}$ for $i = 1, 2, \dots, k$. Then $r_1, r_2, \dots, r_k$ are integers whose product is 1 since $d_{k+1} = d_1$. But that means each ratio r_i equals ± 1 . Moreover $d_1 + d_2 + \dots + d_k = a_k - a_0 = 0$. So, all the d_i 's cannot be of the same sign. So some two consecutive d_i 's are negatives of each other. Suppose $d_{j+1} = -d_j$. Then $d_{j+1} + d_j = 0$. So $a_{j+1} = a_{j-1}$. By cyclicity $a_2 = a_0 = 0$.

Summing up, when every d_i is 0, $a_1 = 0$ and when no d_i is 0, $a_2 = 0$.

An unusual problem (which also appears as Problem A6 in Putnam competition, 2000). Actual examples which represent the two possibilities are $f(x) = x$ and $f(x) = 1 - x$ respectively.

Q.8 Consider the following subsets of the plane:

$$C_1 = \{(x, y) : x > 0, y = \frac{1}{x}\}$$

and

$$C_2 = \{(x, y) : x < 0, y = -1 + \frac{1}{x}\}.$$

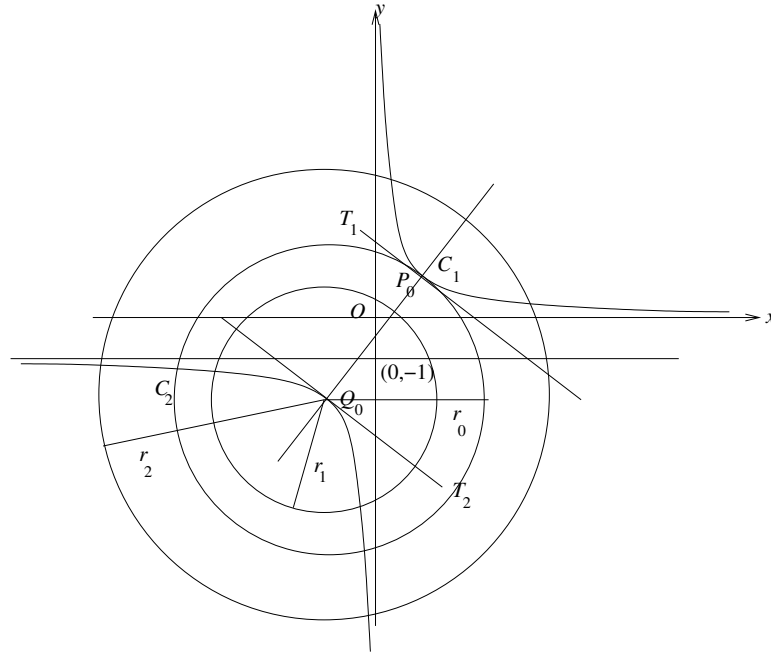
Given two points $P = (x, y)$ and $Q = (u, v)$ of the plane, their distance $d(P, Q)$ is defined by

$$d(P, Q) = \sqrt{(x - u)^2 + (y - v)^2}.$$

Show that there exists a unique choice of points $P_0 \in C_1$ and $Q_0 \in C_2$ such that

$$d(P_0, Q_0) \leq d(P, Q) \text{ for all } P \in C_1 \text{ and } Q \in C_2.$$

Solution and Comments: C_1 is the branch of the hyperbola $xy = 1$ lying in the first quadrant. C_2 is the branch in the third quadrant, but shifted one unit downwards. A typical point P on C_1 is of the form $(x_1, \frac{1}{x_1})$ for some $x_1 > 0$ while a typical point Q on C_2 is of the form $(x_2, -1 + \frac{1}{x_2})$ for some $x_2 < 0$. So $d(P, Q)$ is a function of two variables x_1 and x_2 . Finding the minimum of such a function requires advanced methods. But in the present case the problem can be done geometrically. We recognise that $d(P, Q)$ is simply the distance between the points P and Q . So the problem asks to show that there exists a closest pair of points (P_0, Q_0) with P_0 on C_1 and Q_0 on C_2 . We do so by assuming (tentatively) that such a closest pair exists and then getting clues about its construction.



Consider a circle S_r of radius $r > 0$ centred at Q_0 . Since $Q_0 \notin C_1$, for small values of r such as r_1 , the circle S_{r_1} will not intersect C_1 . On the other hand for r sufficiently large, say $r = r_2$, S_{r_2} will intersect C_1 at two distinct points (as shown in the figure). So there will be some value r_0 of r for which S_{r_0} will meet C_1 in two coinciding points. In other words, it will touch C_1 . For if it intersects C_1 in two distinct points,

then on C_1 there will be some points inside this circle and they will be closer to Q_0 than either of these two points of intersection.

We conclude that if (P_0, Q_0) is the closest pair of points then, the line P_0Q_0 is normal to C_1 at P_0 . By the same reasoning, it is also normal to C_2 at Q_0 .

Thus the search for the closest pair of points reduces to finding x_1, x_2 such that the line joining $(x_1, \frac{1}{x_1})$ and $(x_2, -1 + \frac{1}{x_2})$ is a common normal to both C_1 and C_2 . The equations of the normal to C_1 at $(x_1, \frac{1}{x_1})$ and that to C_2 at $(x_2, -1 + \frac{1}{x_2})$ are

$$y - \frac{1}{x_1} = x_1^2(x - x_1) \quad (1)$$

$$\text{and } y + 1 - \frac{1}{x_2} = x_2^2(x - x_2) \quad (2)$$

respectively. As these two are the same lines, a comparison of coefficients and constant terms gives

$$x_1^2 = x_2^2 \quad (3)$$

$$\text{and } x_1^3 - \frac{1}{x_1} = x_2^3 - \frac{1}{x_2} + 1 \quad (4)$$

Since $x_1 > 0$ while $x_2 < 0$, the first equation gives $x_2 = -x_1$. Putting this into the second equation and simplifying, we get a quartic equation in x_1 , viz.

$$2x_1^4 - x_1 - 2 = 0 \quad (5)$$

Call the L.H.S. as $f(x_1)$. We claim that it has exactly one positive root. The existence of such a root follows from the Intermediate Value Property because $f(1) = -1 < 0$ while $f(2) = 12 > 0$. As for uniqueness, $f'(x_1) = 8x_1^3 - 1$ which has only one zero, viz. $x_1 = \frac{1}{2}$. Hence f has two real roots. Since the complex roots occur in conjugate pairs, their product is positive. But the product of all four roots of $f(x_1)$ is -2 . Hence the product of the two real roots is negative. One of them lies in $(1, 2)$. The other has to be negative. But we are allowing x_1 to take only positive values.

Thus we have proved the existence as well as the uniqueness of the closest pair of points (P_0, Q_0) . ■

Although the existence of a common normal is obvious geometrically, a rigorous derivation demands considerable work (and needs completeness of the real number system, mentioned in the solution to Q.5 above). A problem of a similar type was asked in JEE 2000 where the two curves were parabolas that were reflections of each other in the line $y = x$. (See Exercise (13.23).)

CONCLUDING REMARKS

By and large, Paper 1 problems this year are a lot easier than in the past. Many of them are, in fact, trivial and raise doubts about their inclusion. Certain areas such as inequalities have not been represented at all, while probability has been paid only a lip service. There are hardly any questions about coordinate geometry, except possibly Q.30 which is more about the nature of an equation of a conic. Comparatively, geometry and trigonometry (Q.7, Q.11, Q.12, Q.20, Q.21) have fared better.

Q.13 (about the product of two binomial coefficients being a perfect square) is probably the best question in Paper 1. Q.17 about filling a 4×4 chess board with dominos of certain types) is also a novel one.

Among other good questions are Q.1 (related to Lagrange's interpolation), Q.2 (existence of local extrema of a function with only two critical points), Q.9 (an application of the complex cube root of unity), Q.10 (about possible scores in a test with 20 questions)

Q.5 (a variation of a classic puzzle about a family wanting to cross a river in a row boat), Q.6 (about the possible values of a determinant with only 0 and 1 as possible entries) are good but not novel.

Comparatively, Paper 2 is much better. The first question is more suitable to be asked in Paper 1. Instead a question related to graph theory would have been a better choice. Q.2 is a good problem about evaluating limits of the 1^∞ form. Q.3 asks the image of a region in the complex plane under the squaring function. This has apparently not been asked in the past. Q.4 gives a characterisation of linear functions. But the hypothesis given is stronger than needed. Q.5 about covering a convex quadrilateral is excellent and probably the best question in the entire test. Q.7 is also a novel question about a polynomial function assuming the same values cyclically. Q.8 asks for the existence of a closest pair of points on two curves. It is a good question. But a similar problem has been asked in JEE 2000.

The overall picture one gets is that of a decline in Paper 1 but maintaining good quality in Paper 2.

To conclude, a minor note of approval. There is no problem which has 2019 in it. This practice of including a problem which announces the calendar year of the test, was started probably by the International Mathematics Olympiad. Never mind that in most cases, the particular figure has no significant role. Although originally a novelty, this cheap practice soon became a cliché. It is good that this year the ISI paper setters have come out of it.